

Ziyan Wang_Assignment#6

EC2 instance created

Instance summary for i-0a396fbf456725e31 (ziyan: web-server-instance) [Info](#)

[Connect](#)[Instance state](#) ▼[Actions](#) ▼

Updated less than a minute ago

Instance ID

i-0a396fbf456725e31

IPv6 address

—

Hostname type

IP name: ip-172-31-83-228.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

54.165.78.248 [Public IP]

IAM Role

LabRole [↗](#)

IMDSv2

Required

Public IPv4 address

54.165.78.248 | [open address](#) [↗](#)

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-83-228.ec2.internal

Instance type

t2.micro

VPC ID

vpc-0fd47888347def9 [↗](#)

Subnet ID

subnet-06c78e25b7d48d639 [↗](#)

Instance ARN

arn:aws:ec2:us-east-1:354754152263:instance/i-0a396fbf456725e31

Private IPv4 addresses

172.31.83.228

Public IPv4 DNS

ec2-54-165-78-248.compute-1.amazonaws.com
[open address](#) [↗](#)

Elastic IP addresses

—

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations.

[Learn more](#) [↗](#)

Auto Scaling Group name

—

Managed

false

Go to the terminal. Terraform is installed successfully

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```
[ubuntu@ip-172-31-83-228:~]$ terraform -version
```

Terraform has no command named "-version". Did you mean "version"?

To see all of Terraform's top-level commands, run:

```
terraform -help
```

```
ubuntu@ip-172-31-83-228:~$ terraform version
```

```
Terraform v1.11.2
```

```
on linux_amd64
```

—

Configure the main.tf with requirement in assignment#6

```
.. .bashrc      .profile  .sudo_as_admin_successful  .wget-hsts
ubuntu@ip-172-31-83-228:~$ nano main.tf
ubuntu@ip-172-31-83-228:~$ nano main.tf
ubuntu@ip-172-31-83-228:~$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v5.91.0...
- Installed hashicorp/aws v5.91.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
ubuntu@ip-172-31-83-228:~$ █
```

A screenshot of terminal output showing the public IP address of the EC2 instance ([Screenshot #1]).

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

[Enter a value: yes

aws_security_group.web_sg: Creating...

aws_security_group.web_sg: Creation complete after 2s [id=sg-033d447a18eb64e8b]

aws_instance.web: Creating...

aws_instance.web: Still creating... [10s elapsed]

aws_instance.web: Creation complete after 13s [id=i-0c440fd27a6edc757]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:

instance_public_ip = "54.174.114.209"

ubuntu@ip-172-31-83-228:~\$

Go to the security groups and check the inbound rules attached:

Security Groups (5)

Info

Find resources by attribute or tag

Actions

Export security groups to CSV

Create security group

Name

Security group ID

Security group name

VPC ID

Des

-

[sg-09342cb201ca8b79a](#)

default

[vpc-059994af884d27e1c](#)

defi

-

[sg-0e4a60fe2fa131594](#)

default

[vpc-0fd478888347fdef9](#)

defi

-

[sg-0e424ccb46bd06d4](#)

launch-wizard-1

[vpc-0fd478888347fdef9](#)

laur

-

[sg-033d447a18eb64e8b](#)

web-security-group

[vpc-0fd478888347fdef9](#)

Allc

-

[sg-07ec26411b8811e50](#)

Ec2SecurityGroup

[vpc-059994af884d27e1c](#)

VPC

A screenshot from the AWS Management Console showing the security group configuration ([Screenshot #2])

sg-033d447a18eb64e8b - web-security-group

[Actions ▾](#)

Details

Security group name
web-security-group

Security group ID
sg-033d447a18eb64e8b

Description
Allow inbound HTTP traffic from anywhere

VPC ID
vpc-0fd47888347fdef9

Owner
354754152263

Inbound rules count
2 Permission entries

Outbound rules count
1 Permission entry

[Inbound rules](#)[Outbound rules](#)[Sharing - new](#)[VPC associations - new](#)[Tags](#)

Inbound rules (2)

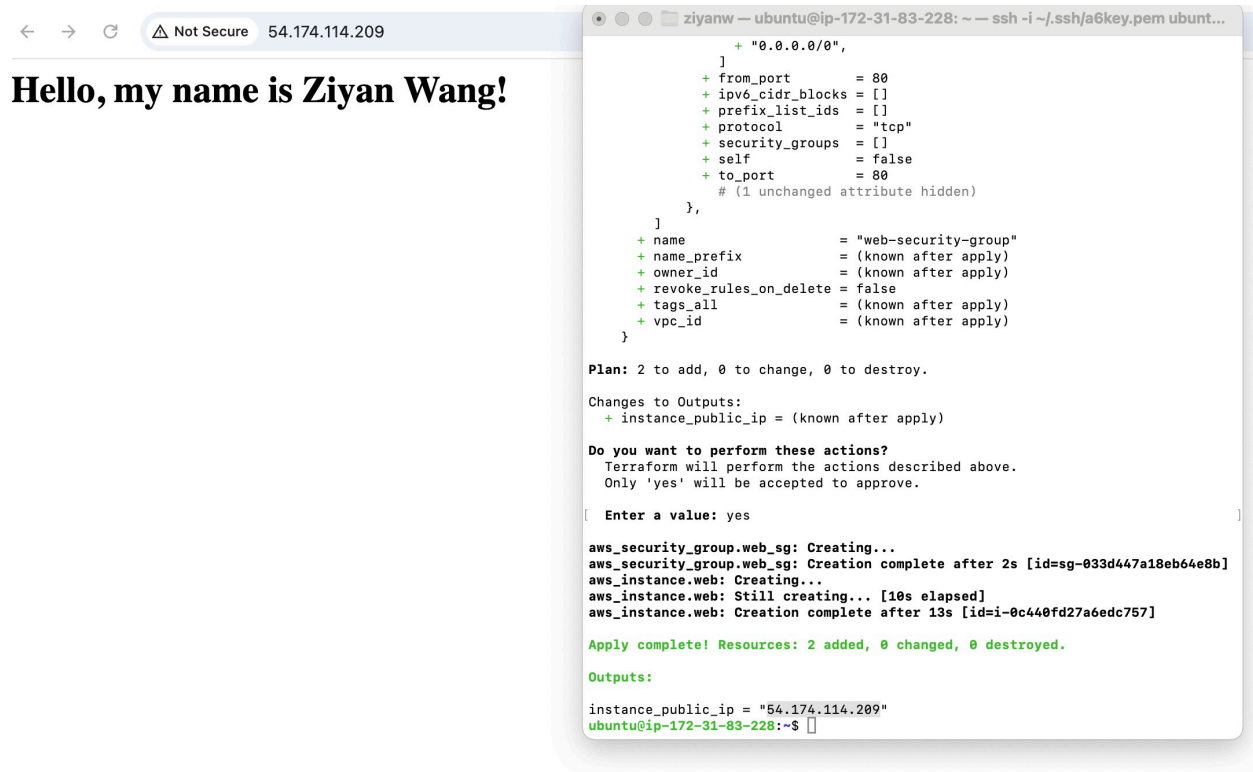
[Manage tags](#)[Edit inbound rules](#)[< 1 >](#)

☐	Name ▾	Security group rule ID ▾	IP version ▾	Type ▾	Protocol ▾	Port range
☐	-	sgr-0387b541008290e09	IPv4	HTTP	TCP	80
☐	-	sgr-0f6462cd61819160b	IPv4	SSH	TCP	22

A screenshot showing the custom index.html page served on the public IP of your EC2 instance via Nginx in the web browser ([Screenshot #3]).

← → ↻ Not Secure 54.174.114.209

Hello, my name is Ziyan Wang!



Using Curl + ip also works:

```
ubuntu@ip-172-31-83-228:~$ curl http://54.174.114.209
<h1>Hello, my name is Ziyan Wang!</h1>
ubuntu@ip-172-31-83-228:~$
```

Check all the files are exist:

```
ubuntu@ip-172-31-83-228:~$ nano variables.tf
ubuntu@ip-172-31-83-228:~$ nano outputs.tf
ubuntu@ip-172-31-83-228:~$ nano main.tf
ubuntu@ip-172-31-83-228:~$ ls -l
total 24
-rw-rw-r-- 1 ubuntu ubuntu 1402 Mar 20 04:20 main.tf
-rw-rw-r-- 1 ubuntu ubuntu 121 Mar 20 04:18 outputs.tf
-rw-rw-r-- 1 ubuntu ubuntu 10652 Mar 20 04:06 terraform.tfstate
-rw-rw-r-- 1 ubuntu ubuntu 458 Mar 20 04:18 variables.tf
ubuntu@ip-172-31-83-228:~$
```

A screenshot of the terminal output from terraform destroy, confirming successful deletion of all resources. ([Screenshot #4]).

Plan: 0 to add, 0 to change, 2 to destroy.

Changes to Outputs:

- instance_public_ip = "54.174.114.209" -> null

Do you really want to destroy all resources?

Terraform will destroy all your managed infrastructure, as shown above.
There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

```
aws_instance.web: Destroying... [id=i-0c440fd27a6edc757]
aws_instance.web: Still destroying... [id=i-0c440fd27a6edc757, 10s elapsed]
aws_instance.web: Still destroying... [id=i-0c440fd27a6edc757, 20s elapsed]
aws_instance.web: Still destroying... [id=i-0c440fd27a6edc757, 30s elapsed]
aws_instance.web: Still destroying... [id=i-0c440fd27a6edc757, 40s elapsed]
aws_instance.web: Destruction complete after 40s
aws_security_group.web_sg: Destroying... [id=sg-033d447a18eb64e8b]
aws_security_group.web_sg: Destruction complete after 1s
```

Destroy complete! Resources: 2 destroyed.

```
ubuntu@ip-172-31-83-228:~$
```
